

Seunghyun Im

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MComp student in **Artificial Intelligence and Computer Science** at the **University of Sheffield** with experience in machine learning, medical imaging analysis, and computer vision. Skilled in Python, PyTorch, and data analysis with research experience at the Insigneo Institute and multiple AI-based projects involving deep learning and computer vision. Have a strong interest in solving complex problems.

EDUCATION

MComp Artificial Intelligence and Computer Science <i>University of Sheffield</i> Integrated Masters : 2:1	09/2020 – 07/2026 Sheffield, United Kingdom
KIS Diploma <i>Kodaikanal International School</i>	07/2013 – 06/2019 Kodaikanal, India

PROFESSIONAL EXPERIENCE

Summer Research Placement <i>Insigneo Institute</i> - Developed a unified sensor data harmonisation pipeline integrating 4 heterogeneous wearable IMU datasets into a standardised 2.56 s sliding-window benchmark for 6-axis lower-back accelerometer/gyroscope signals. - Designed a JSON-config preprocessing architecture enabling plug-and-play integration of future clinical cohorts (Parkinson's, CP, PSP) without codebase modification. - Training baseline ML/DL activity classifiers (Random Forest, XGBoost, CNN/LSTM) on healthy participant data to establish transfer-learning benchmarks for patient populations. Tech stack: Python, PyTorch, pandas, NumPy, scikit-learn, signal processing, JSON, HPC, Machine Learning Algorithms.	06/2026 – Present Sheffield, United Kingdom
Faculty Rep <i>University of Sheffield</i> Represented the students' experience regarding modules during meetings with the Computer Science department and the Student Union.	10/2025 – 06/2026 Sheffield, United Kingdom
UG Demonstrator <i>University of Sheffield</i> Facilitated first-year module tutorial sessions to support students in understanding Ruby.	09/2025 – 06/2026 Sheffield
Summer Research Placement <i>Insigneo Institute</i> - Insigneo Institute is a high-level research initiative in partnership with the University of Sheffield and the NHS. - Developed Python-based pipelines to analyse MRI and CT lung imaging datasets in an HPC environment. - Implemented algorithms to classify lung conditions (asthma, COPD, healthy) using medical image processing techniques. - Co-authored a research poster presenting the project outcomes.	06/2025 – 09/2025 Sheffield, United Kingdom
Co-Secretary <i>Korea Society of University of Sheffield</i> - Secured sponsorships by liaising with multiple companies and organizations. - Successfully onboarded Jongga as a sponsor for Sheffield Korea Day, enhancing event visibility and credibility.	09/2024 – 09/2025 Sheffield, United Kingdom
Interpreter <i>ConnectorS</i> Provided on-site English–Korean interpretation during a business meeting with Sheffield Hallam University.	09/2024 – 09/2024 Sheffield, United Kingdom
Sergeant <i>Republic of Korea Army</i> - Maintained and repaired military communication equipment, including PRC-999K radios and TMR systems. - Led a squad as a Sergeant for 4 months, coordinating equipment management and operational readiness.	09/2021 – 03/2023 Seoul, Korea, Republic of

PROJECTS

Seunghyun Im Portfolio Github

Summer Research Placement 06/2026 – Present

Wearable HAR Data Harmonisation Pipeline

- Architected an end-to-end ETL system processing 6.4M+ raw IMU samples from 4 datasets into ML-ready tensors (N, 128, 6) and statistical feature vectors (N, 54).
- Implemented stratified train/val/test splits with persisted StandardScaler artifacts for reproducible model training.

Retail Demand-Driven Recommendation System 06/2026 – Present

Personal Project

- AI-powered retail recommendation system that predicts what customers will buy next, integrated with demand forecasting to maximize sales and minimize waste.
- Completed: Database architecture, ETL pipeline, business EDA with SQL Window Functions
- In Progress: Neural Collaborative Filtering, Two-Tower Model, FastAPI serving

Turing Project 09/2025 – Present

Geospatial Data Correction Project

- Built an automated spatial realignment pipeline (Python, PyTorch) for UAV forestry datasets
- Segmented tree crowns using U-Net and exported geometries to QGIS for validation
- Applied LE-JEPA for self-supervised feature extraction (128D embeddings)
- Developed a sliding-grid algorithm with Random Forest & DBSCAN for coordinate correction
- Scaled solution to handle 17,000+ high-complexity data points

Summer Research Placement 06/2025 – 10/2025

Contrast agent free regional ventilation imaging in CT and MRI

- Developed a Quadtree algorithm for MRI lung image analysis using Python and SimpleITK.
- Evaluated 2D and 3D models using Pearson correlation, MSE, and MAE metrics.
- Generated Quadtree representations from CT images and quantified heterogeneity using specific gas volume metrics.
- Analysed the outcomes to distinguish between three lung conditions (asthma, COPD, and healthy).

Dissertation Project 06/2024 – 06/2025

Sports video classification and summarisation

- Developed a sports video classification model using CNN + LSTM, achieving 94% accuracy on the UCF-101 dataset.
- Built two GUI applications for video classification and highlight summarisation.

COURSES

AI-SW Education 06/2022 – 05/2023

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Remote, South Korea

- Completed sessions on AI, data analysis, and machine learning.

SKILLS

Technical Skills

- Languages: C#, C++, Java, Python, Ruby
- Frameworks: PyTorch, OpenCV, ROS
- Tools: Git, Unity, MySQL, Excel

Graphical Skills

- Adobe Premiere Pro, Illustrator, Photoshop, Maya, CAD

Language Skills

- Korean
- English

Soft Skills

- Leadership, Team Management, Project Planning, Agile Development, Data Analysis, Problem Solving

AWARDS

IBM prize - Runners Up 2024 2024

IBM and University of Sheffield

- Runner-Up of the Year by IBM and the University of Sheffield.

The Global Engineering Challenge 2021 27/03/2021

University of Sheffield

- Best Communicated Solution, Best Overall Solution, Best Report.